

Disaster Mangement Robot Using YOLO V5 for Detection and ROS Noetic For Robot Operation



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Dedicated to my family especially my mother for her
unconditional love, care and continuous encouragement
throughout my life.

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Abstract

In the wake of earthquakes and other natural disasters, timely and effective rescue operations are crucial to saving lives. Delays in response can lead to significant casualties, underscoring the need for advanced technological solutions. This work presents the deployment of deep learning algorithm to detect and classify injured humans in the event of disaster which allows targeted and rescue operations. The robotic car is embedded with a Jetson Xavier (8-core NVIDIA Carmel Arm®v8.2 64-bit CPU), PID-controlled motors, camera for image capture, and a robotic arm. Utilizing deep learning algorithms, the system can detect and classify injured humans in disaster scenarios, allowing for targeted and efficient rescue actions. Furthermore, this work integrates Lidar technology for precise navigation, obstacle detection, and avoidance. The proposed solution aims to improve assessment of disaster impact, addressing the critical need for efficient and reliable disaster response mechanisms. The test results demonstrate that our proposed and developed Object Detection algorithm achieves precision and recall of 87 and 85 percent, respectively. We also observe the decrease in the losses over epochs making the model more suitable to adopt to different disaster scenarios.

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Chapter 1

Introduction

1.1 Background

A significant increase in natural disasters has been observed in recent years. Disasters like earthquakes, floods, blasts, and many other disasters cause large damage and are therefore a threat to human lives. In practice disaster management systems rely mainly on humans for search and rescue operations. However, it's not possible for humans to quickly reach out for help in the affected areas, which can result in the loss of precious lives and long-suffering for the injured ones.

1.2 Problem Statement

Despite the fact that disaster management technologies have advanced significantly over the year, there are still large gaps in these system's ability to precisely locate injured people in affected locations. Human rescuers face serious challenges, including unstable and heavy building structures and various risky situations like fires and chemical exposure. To avoid delay in rescue efforts the disaster management robot can search and locate injured people in a difficult setting and rescue them therefore improving the reaction time and reducing casualties.

1.3 Objectives

The primary goal of this project is to design and build a disaster management robot that can move autonomously into disaster-affected areas. We use the YOLOv5 algorithm to identify injured humans in real-time, it also aims at incorporating a camera and Lidar for environment perception and obstacle sensing respectively. Furthermore to control the robot's movements and to make it stable we used PID-Controlled motors.

1.4 Project's Scope

This project will involve the assembling of a disaster management robot that is capable of object detection and autonomous navigation. Components such as a Jetson Xavier, a camera, servo motors for controlling the robotic arm, lidar for creating a map, and PID-controlled motors for navigation will be incorporated into the making of the robot. The project will integrate these components to make a robot fit for search and rescue tasks in disaster-stricken zones. Nonetheless, this project is limited by factors like a predetermined budget for the project, accessibility to the proper and relevant parts of hardware, and real-life disasters.

1.5 Thesis Structure

This thesis is divided into sub chapters, each having information about different aspects of the project

Chapter 1: Introduction - This section summarizes the project's origins, the main issue it aims to solve, its goals, its focal point, and the way this thesis is structured.

Chapter 2: Literature Review - This section examines the body of research on robots for disaster relief, object recognition, navigation, and other closely related fields.

Chapter 3: Methodology – These sections explain the full details about the

robot consist of the machine, system, and the system controlling the robot.

Chapter 4: Implementation – describe one of the phases of development that involve the composition of the required components and their subsequent check.

Chapter 5: This chapter presents the results obtained from experiments conducted on the disaster management robot under various conditions, followed by a discussion of the findings. The results are evaluated against the initial objectives of the project, while also addressing the limitations encountered during development and testing. Finally, recommendations for future improvements are identified.

Chapter 6: Conclusion – Presents the main results of the project work and specifies further research prospects.

Chapter 2

Literature Review

Disaster management is one of the fields that have recorded progress in the last few years because of robotic and AI integration. Such technologies are expected to shrink search time and risks involved in search and rescue operations. This chapter provides an overview of the previous studies on disaster management robots, methods of object detection and navigation, and points out the deficiencies that this project intends to fill.

2.1 Disaster Management Robots

Automation particularly the use of robots in disaster management has been researched for at least more than 10 years. Likewise, Murphy *et al.* (2008) explained that robots are very useful in search and rescue operations for the reason that it is dangerous to send human beings into unstable structures, and even in disasters where people are trapped, it is not advisable to endanger their lives in the process of rescuing them. Compact robots such as PackBot and Talon have been utilized in instances including the 9/11 World Trade Center collapse as well as the Fukushima Daiichi nuclear disaster.[1] There are many investigations and innovations regarding the structuring and implementation of such robots for managing disasters. The authors Murphy et al.[2] describe the distinction of utilizing robots in responding to a disaster and the problems and advantages of employing robots in a search and rescue operation. Simi-

larly, the paper by Casper and Murphy [3] describes the actual use of robots after the World Trade Center 9/11 experience and how effective they are or not in real problems.

Analyzing the work of Kitano et al. [4], the authors discuss the application of the robotic teams within the simulated disaster scenarios that proved that multi-robot communication and coordination is sufficient and obligatory for the accomplishment of search and rescue tasks. Another valuable piece is the paper by Tadokoro et al. [5], where the authors describe the creation of rescue robots that could operate in ruins and find victims with the help of innovative equipment.

However, these robots have a lot of limitations about automatism of motion and decision-making in time. A disaster environment is quite challenging and thus a need for a robot that can operate on its own, perceive its surroundings, and make decisions on its own to search for and rescue a trapped person.

2.2 Object Detection Algorithm

To decrease the number of incidents that happen during the process of people rescuing who are stuck in the debris, object detection is a critical component in the construction of autonomous rescue robots. Redmon et al. (2016) introduced the YOLO algorithm, which can be considered as one of the most accurate and the fastest methods of real-time object detection. Consequently, due to its superior detection efficiency of objects, throughput rate, and low latency compared to the original method, YOLOv5 can be described as the development of the method (Jocher et al. , 2020).

Some examples in which the YOLO network has been applied include robotics, traffic monitoring, and surveillance. Although there are studies on the deployment of this YOLOv5 network, it has never been opted for the disaster management, particularly in the search of victims. This work fills the afore mentioned gap by applying YOLOv5 in real-time injury detection in disaster area.[6]

2.2.1 YOLOv5 Framework

YOLOv5 (You Only Look Once, Version 5)[7] is a state-of-the-art object detection algorithm known for its speed and accuracy, making it highly applicable in real-time applications. When the system is given a frame, several critical steps describe the process it will go through:

- **Input Frame Processing:** The input frame, typically an RGB image, is first resized to a fixed dimension (usually 640x640 pixels). This resizing ensures consistency across various input sizes, facilitating faster and more reliable processing.
- **Feature Extraction:** The resized frame is then passed through a series of convolutional layers, which serve as feature extractors. These layers detect various features such as edges, textures, and patterns that are crucial for identifying objects within the frame.
- **Object Detection:** YOLOv5 employs a single neural network to predict bounding boxes and class probabilities directly from full images in one evaluation. This network divides the image into a grid, and for each grid cell, it predicts multiple bounding boxes and their corresponding confidence scores. The confidence score reflects the likelihood that a bounding box contains an object and the accuracy of the bounding box prediction.
- **Non-Maximum Suppression (NMS):** To refine the detection results, YOLOv5 applies non-maximum suppression, which eliminates redundant bounding boxes. This step ensures that each detected object is represented by only one bounding box, minimizing false positives and improving detection accuracy.
- **Output:** The final output consists of bounding boxes around detected objects, along with their class labels and confidence scores. This information is crucial for the robotic car to identify and locate humans in disaster scenarios, enabling it to take appropriate rescue actions.

2.3 Autonomous Navigation and Sensor Integration

For robots helping in disaster management, it is critical to incorporate autonomous navigation, helping robots to move in a complex and dynamic environment. The navigation includes Lidar for environment mapping and cameras together with other sensors to detect obstacles. In a study by Zhang et al. (2019), the authors used lidar and visual sensors for simultaneous localization and mapping (SLAM) in robotics. The authors established the fact that environment can be accurately mapped in real time.[8]

Furthermore, the incorporation of PID-controlled motors can improve the robustness and accuracy of robotic applications. For holding the desired position and velocity of motors [9] used by robots, PID controllers have played a significant role (Åström and Murray, 2008). In our work we utilised above mentioned methods in order to come up with a good navigation system for disaster management robot.

2.4 Gaps in Current Research

1. Real-time Injury Detection: In existing studies YOLO and other object detection algorithms have been explored in multiple contexts. These approaches never recieved attention to identify injured people in emergencies.
2. The field of autonomous navigation in highly dynamic and unstructured disaster scenarios is still in its early stages of development. This is especially true for the integration of modern sensors with autonomous navigation algorithms.
3. Comprehensive System Integration: For efficient disaster management, a comprehensive system integrating robotic manipulation, autonomous navigation, and object identification is required.

2.5 Conclusion

This project aims to address these research and development gaps discussed in Section 2.4 by developing a disaster management robot equipped with YOLOv5

for real-time detection of injured individuals, autonomous navigation capabilities using lidar, camera sensors, and PID-controlled motors for precise movement. The integration of these components will create a robust system capable of enhancing search and rescue operations in disaster-affected areas.

Chapter 3

Methodology

This chapter presents the interfacing of multiple sensors and development of an application for the disaster management robot. This refers to the establishment of the system and its components, including the system's hardware and software, data augmentation and pre-processing, the training of the model on the custom dataset, and the overall interfacing of the system. The chapter also describe the methods, tools, algorithms and libraries in addition to the mode of data collection and analysis processes.

3.1 System Architecture

The disaster management robot is an autonomous system which can identify and move the injured people from the disaster affected area. In this section, we elaborate the architecture of proposed autonomous robot to help rescue people in the event of disaster . The propose rescue robot is capable to rescue people where conventional emergency system fails to reach.

3.1.1 Overview

The robot is equipped with a camera and a trained YOLOv5 model for disaster detection. Upon identifying a disaster, the robot autonomously navigates towards the affected area. The YOLOv5 model is also trained to recognize injured individuals. When an injured person is detected, the robot carefully lifts

them using its robotic arm and initiates a rescue operation. Once the rescue is completed, the robot returns to its initial position and safely places the injured person down.

The detection and identification processes are managed by the YOLOv5 model, which has been trained on a custom dataset encompassing three classes: Disaster, Alive, and Injured. The model extracts these classes from the camera feed, guiding the robot's subsequent actions.

Key components of the robot include a camera that captures real-time video, which is processed by the Jetson Xavier to identify objects and control navigation. A lidar sensor provides environmental data to facilitate obstacle avoidance. The robotic arm, actuated by servo motors, is responsible for lifting and transporting injured individuals. Finally, PID-controlled motors ensure precise movement and accurate directional control.

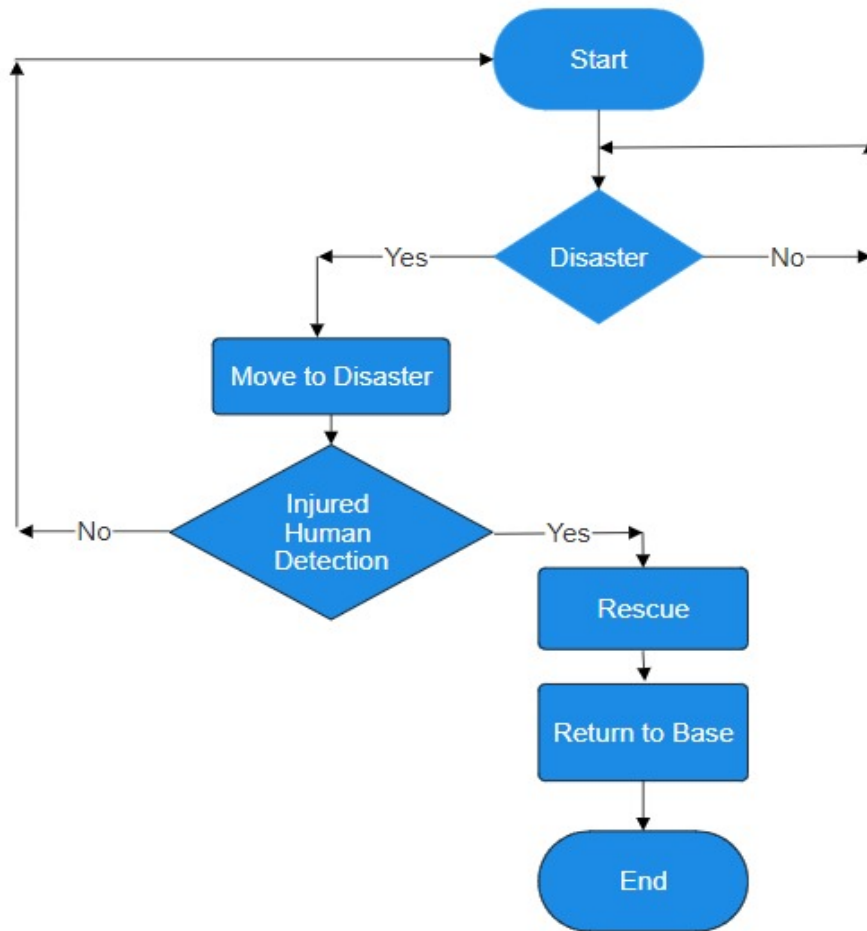


Figure 3.1: Overview Of How The Robot Work

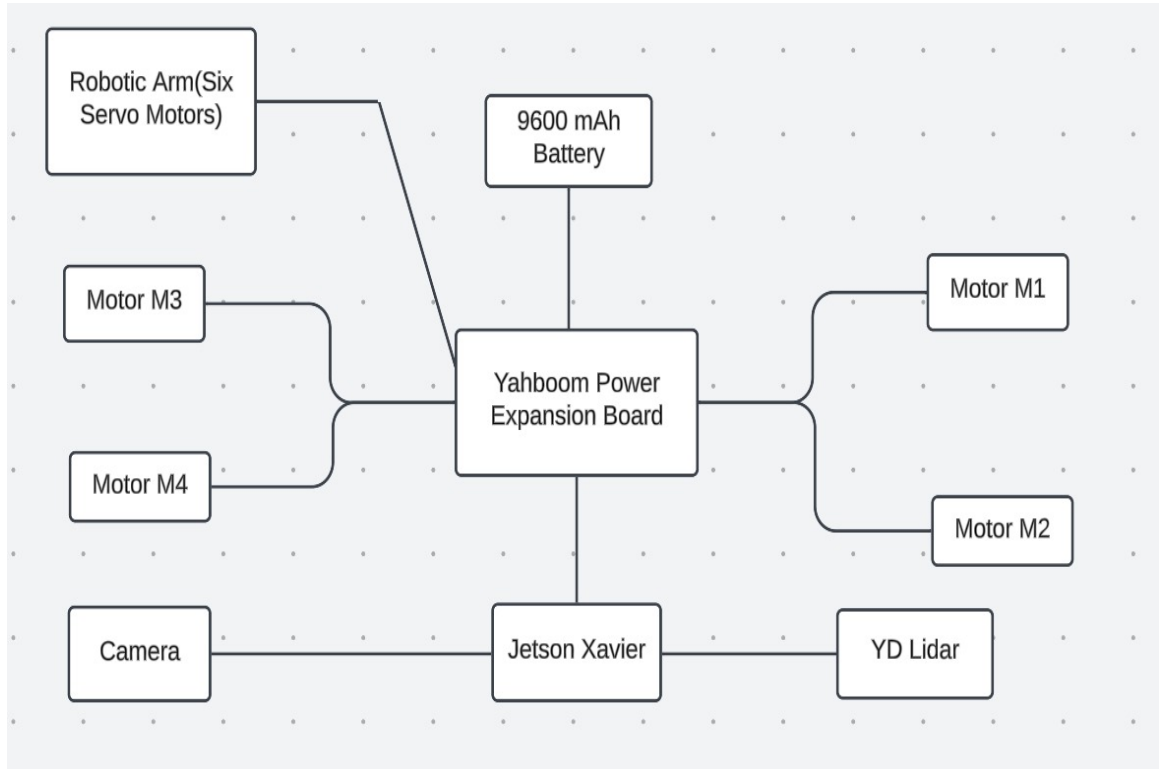


Figure 3.2: System Architecture

3.2 Hardware Components

3.2.1 Jetson Xavier

NVIDIA's Jetson Xavier is an advanced AI computing system that is used by this robot to handle all types of computations. It features an eight-core ARM v8 CPU and a powerful 512-core Volta GPU with Tensor Cores, making it highly efficient for running deep learning models and handling complex computations. This system supports high-speed data processing and allows for the implementation of sophisticated AI algorithms, which are essential for tasks involving detection. (NVIDIA, 2019).

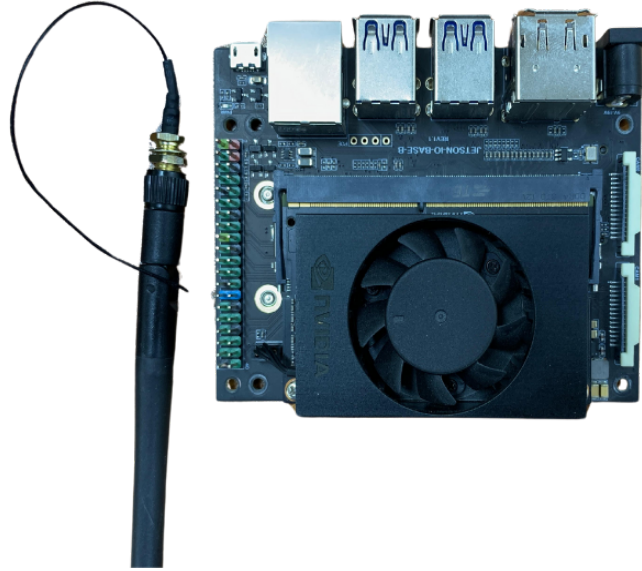


Figure 3.3: Jetson Xavier

3.2.2 Camera

The Astra Pro Plus camera is used for object detection to help the robot identify injured people. With a range of 0.6 meters to 8 meters and a field of view of 60° horizontal by 49.5° vertical by 73° diagonal, the Astra Pro Plus is optimized for long-range use cases. Its RGB image resolution of 1920 x 1080 at 30 frames per second and depth image resolution of 640 x 480 at 30 frames per second ensures high-quality imaging. The camera's compact size of 165mm x 30mm x 40mm and operational temperature range of 10 to 40 degrees Celsius make it versatile and easy to integrate into various environments.



Figure 3.4: Astra Plus Pro Camera

3.2.3 Servo Motors

The robotic arm is powered by high-torque serial bus smart servos motors. The servo motors are used in the robotic arm to help the robot lift the injured humans. The servo motor provides 15 kg·cm of torque and is built with durable metal gears, ideal for tough tasks. It offers features like angle reading, speed control, and ID changes, which can be managed via comport commands and daisy-chained for complex operations. The motor is also easily controlled using the Yahboom servo driver board.



Figure 3.5: Servo Motor

3.2.4 Robotic Arm

The robotic arm is used by robot in lifting injured human. It's made of an aluminum alloy bracket and a strong structure of a chassis. It includes Six bus servo motor and a 3MP camera. This arm has robust construction, and high resolution movements, which helps it lift heavy objects.



Figure 3.6: Robotic Arm

3.2.5 DC Motors

The robot utilizes 520 DC gear motors with encoders, that will help the robot move with various speeds to suit different operational needs. These speeds are 205 RPM, 333 RPM, and 550 RPM. These motors help in the suitable handling of motions and directions of the robot.



Figure 3.7: DC Motor

3.2.6 Lidar Sensor

The robot features the YDLIDAR 4ROS, for environment mapping and obstacle avoidance. It has a range suitable for wastewater pipes up to 30 meters in length and a high sampling frequency of 2000 times per second. The sensor is designed to handle bright environments with up to 100KLux, allowing for effective ranging and mapping both indoors and outdoors. Additionally, it is impact-resistant and boasts an IP65 rating, ensuring reliable performance in waterproof and dustproof conditions.



Figure 3.8: 4ROS YD LIDAR

The lidar sensor utilizes TOF[Time-of-Flight] technology for Environmental perception and identification of obstacles.



Figure 3.9: Hardware Components Of The Robot.

3.3 Integration of Components

3.3.1 Power Supply And Connectivity

The robot hardware parts are driven by a 12V 9600 mAh lithium-ion battery. This battery is enough to allow the functioning of all integrated elements so that the robot can effectively perform tasks in disaster situations. The hardware components are connected through the USB expansion boards which creates a reliable interface to link most of the parts of the system together.

3.3.2 Transferring and Processing of Data

The camera feed the video streams of the environment to Jetson Xavier where they are converted to frames using OpenCV. These video frames are processed by the YOLOv5 object detection model, running on the Jetson Xavier, for the identification of injured. The results of detection are marked and passed to the control system for further regulating the robot's functioning.

3.3.3 Control Mechanisms

By the detection labels generated by YOLOv5, the control system drives the communication with the DC and servo motors to execute appropriate actions. For example, when a disaster is detected the robot moves to the area where disaster is detected. Once it had reached there it started looking for injured individuals. Once an injured human miniature is detected, the robotic arm with the help of the servo motors lifts the individual and moves him/her to a safer area.

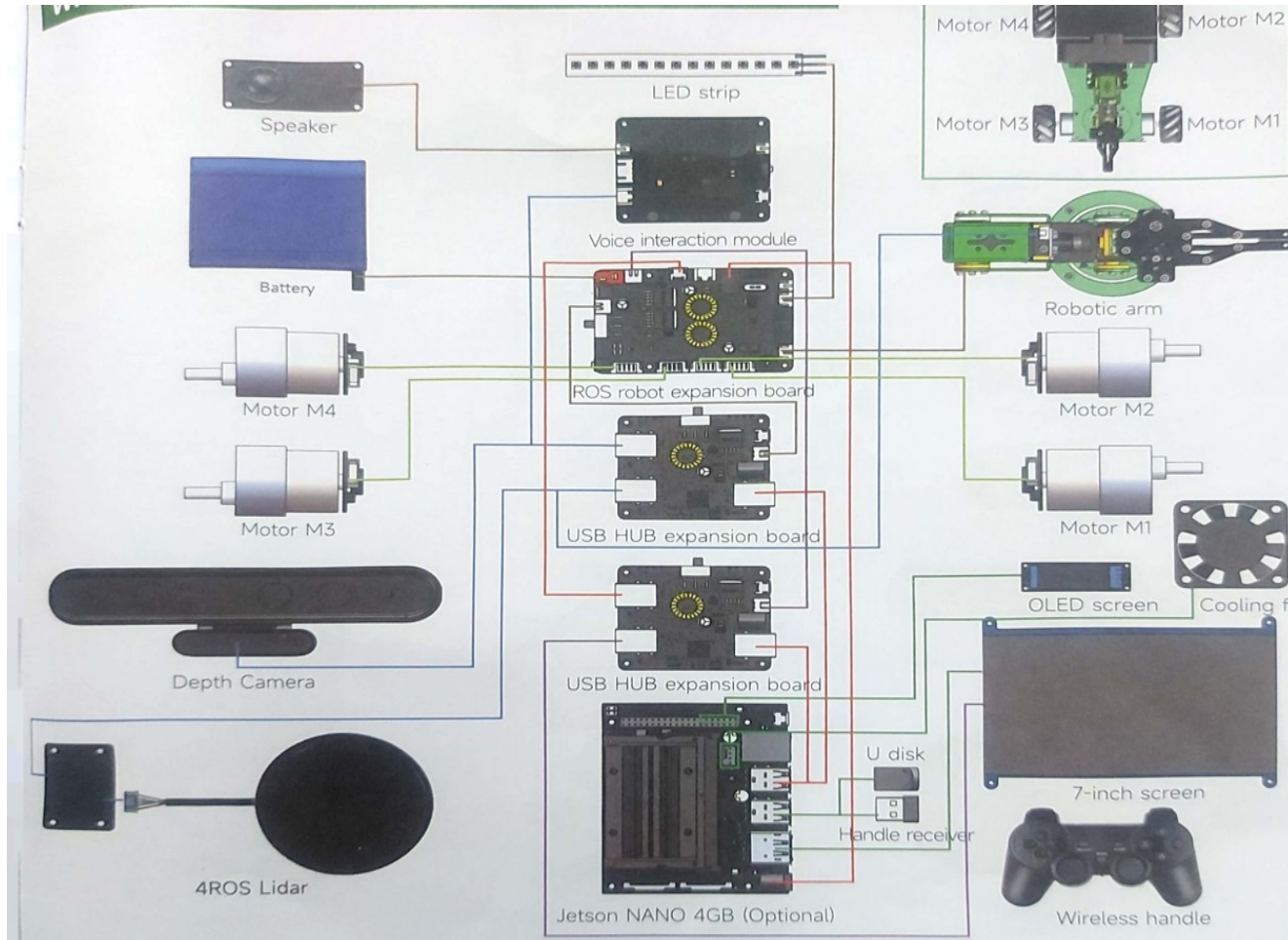


Figure 3.10: Integration Of Components

3.4 Object Detection and Navigation

3.4.1 YOLOv5 Training and Optimization

We have developed a custom dataset for the application that fits the disaster management domain. The dataset comprises three classes: disaster, alive, and injured. The system was built by simulating some disaster cases and getting some videos from different angles. The collected videos were extracted frame-wise with 5 fps by a Python script, where the frames were sampled. The

images were manually annotated with the help of the Labelling tool, which is a graphical image annotation tool that allows accurate marking of objects in the frames.

With the dataset prepared, the fifth version of the You Only Look Once algorithm, YOLOv5, was then trained. YOLOv5 is one of the most advanced and powerful approaches for conducting real-time object detection, making it highly accurate and fast. Based on this, hyper-parameters tuning was undertaken most optimally to ensure improved model performance while achieving the set objectives of classifying classes within our disaster scenarios.

3.4.2 Navigation and Obstacle Avoidance Algorithms

For the location of the obstacles, the robot uses a lidar sensor; which is the YDL-IDAR 4ROS in this case. This lidar sensor employs the TOF (Time-of-Flight) method that when applied produces a comprehensive SLAM (Simultaneous Localization and Mapping) 'map' of the surrounding. Localization is made possible in the SLAM process in which the robot can interpret its environment and its position on the map.

3.4.3 Decision-Making for Movement and Interaction

The YOLOv5 detections influence the robot's decision-making process of movement and interaction. During initial detection, the robot scans its surroundings for signs that a disaster has taken place. If there is a disaster label, the robot navigates to the area as indicated. Once at the site of the disaster, YOLOv5 carries out additional processing to find any injured people on the scene. When an injured human is detected, the robot navigates towards the injured person, carefully picks it up with the robotic arm, and transports it back to the base location.

3.5 Testing and Validation

3.5.1 Testing Environments And Evaluation

For assessing the proficiency and stability of the disaster management robot, the testing included a well-created physical environment, which was set up to replicate disaster conditions perfectly. We used thermopole to create 4 buildings, out of which one was brought down intentionally. Actual objects were placed to mimic a disaster scenario. Moreover, stones were put around the table. Additionally, we placed miniature human figures in the setup some are standing to represent alive people and those on the ground with red ink to represent the victims. The primary metric used to evaluate the system's performance was the precision and recall which were 87 percent and 85 percent respectively.

3.5.2 Specific Tests and Outcomes

We conducted several specific tests to validate the effectiveness of our system. The first test involved the initial disaster scenario described above. The robot successfully navigated through the debris, identified the disaster site, detected the injured individuals, and performed the rescue operation as intended. The high accuracy of YOLOv5 was evident in this test, with a few minor misclassifications that were subsequently addressed through further training and optimization.

A second, distinct disaster scenario was created to further evaluate the model's robustness and adaptability. This scenario included different configurations of debris and varied positions of the miniature figures. Although the model made some mistakes, such as misclassifying alive individuals as injured, the overall performance was satisfactory, demonstrating the system's ability to adapt to new environments and maintain a high level of accuracy.

Chapter 4

Implementation

4.1 Training the YOLOv5 Model

4.1.1 Dataset Collection and Annotation

For the training of the YOLOv5 network, a custom dataset was prepared with an emphasis on identifying injured persons in disaster-affected regions. There is a total of 5154 images in which the training folder contains 3607 images while the images in the validation folder are 1547 in total. The following images were taken from a disaster-based scenario created using a destroyed thermopole building along with stones. Human figures were put inside the scene to depict alive and injured individuals.

Labellmg application was used for image annotation, there was strict compliance with the required guidelines for bounding the objects of interest. The annotations included three classes:

- **Alive (class 0):** Represented by human miniatures that are not injured.
- **Injured (class 1):** Represented by human miniatures that appear to be injured.
- **Disaster (class 2):** Represented by the disaster elements in the scene, such as debris and the destroyed building.

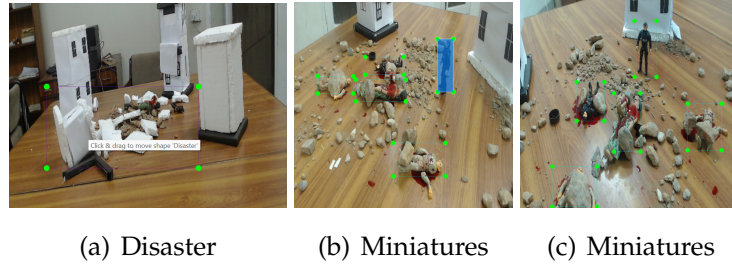


Figure 4.1: Dataset Annotation.



Figure 4.2: Injured Class



Figure 4.3: Alive Class

4.1.2 Training Process and Parameters

The YOLOv5 model was trained using a custom dataset using the below parameters and steps. The images were preprocessed to meet the input requirements of the YOLOv5 architecture by resizing them to 640 x 640 pixels. The

model configuration was then adjusted by modifying the classes in the YOLOv5 configuration file to match the custom classes: Alive, Injured, and Disaster. Additionally, the paths for the training and validation folders were updated with the respective dataset paths. To improve training and reduce overfitting, adjustments were made to the learning rate. The best performance was achieved after a total of 79 epochs. The entire training process was carefully conducted to ensure optimal results.

4.1.3 Control Logic and Algorithms

The control logic for the disaster management robot was implemented in Python, using OpenCV and ROS. The YOLOv5 algorithm was used to detect injured people in the video stream obtained using the camera. The detected objects were identified and given labels based on confidence of detection. The lidar sensor data collected was utilized in creating the SLAM map that would help the robot to reach the disaster while avoiding obstacles. The position and the velocity of the robot depended on the results of the objects' detection. Thus, it moved to areas where disaster was detected, and the action was to grab injured people with the help of the robotic arm. The control signals for the DC motors and servo motors that were used in the robots were derived from the position and orientation of the detected object.

Chapter 5

Testing and Results

5.1 Experimental Setup And Performance Metrics

The dataset is divided into training and validation sets. Performance metrics and losses are calculated at each epoch. Box loss, objectness loss, classification loss for both training and validation sets, precision, recall, and mAP values.

- **Precision:** It answers the question "Out of all the instances that the model predicted as positive, how many were positive?". High precision indicates that the model has a low false positive rate.
- **Recall:** It answers the question "Out of all the actual positive instances, how many did the model correctly identify?". High recall indicates that the model has a low false negative rate.
- **Training Box Loss:** Training Box Loss refers to the error between the predicted bounding boxes and the actual ground-truth bounding boxes during the training process.
- **Training objectness Loss:** Refers to the error or discrepancy between the predicted objectness scores (probabilities that a specific region contains an object) and the actual ground-truth labels (indicating whether the region contains an object or is just background) during the training process.

- **Training Classification Loss:** Refers to the error or discrepancy between the predicted class labels for objects within the bounding boxes and the actual ground-truth class labels during the training process.
- **Validation Box Loss:** Validation Box Loss refers to the error between the predicted bounding boxes and the actual ground-truth bounding boxes during the model testing on validation data.
- **Validation objectness Loss:** Refers to the error or discrepancy between the predicted objectness scores (probabilities that a specific region contains an object) and the actual ground-truth labels (indicating whether the region contains an object or is just background) during the model testing on validation data.
- **Validation Classification Loss:** Refers to the error or discrepancy between the predicted class labels for objects within the bounding boxes and the actual ground-truth class labels during the model testing on validation data

5.2 Results Analysis

Significant improvements were observed in the precision and recall over epochs. The mAP metrics showed a steady increase indicating better detection capabilities. A Decreasing trend was observed in both training and validation losses showing model improvements and generalization.

TESTING AND RESULTS

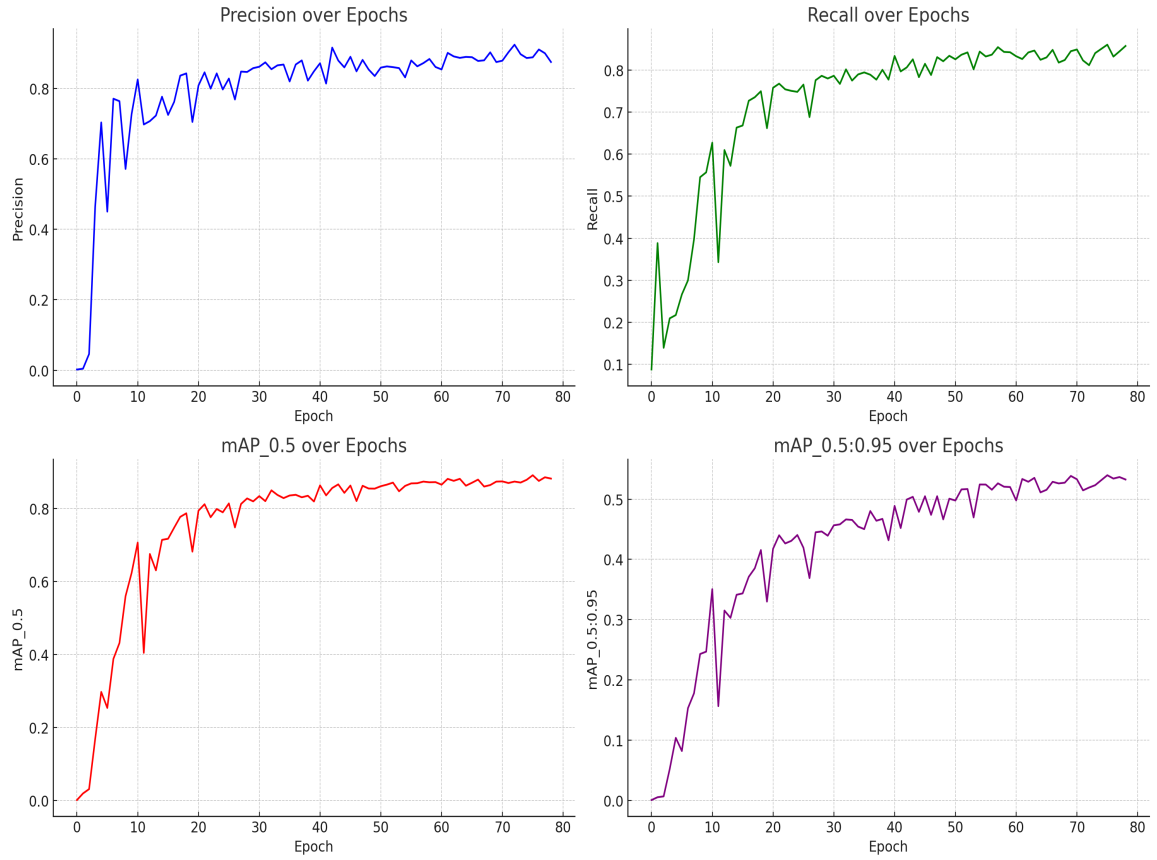


Figure 5.1: Precision, Recall, mAP 0.5 and mAP 0.5-0.9 over Epochs

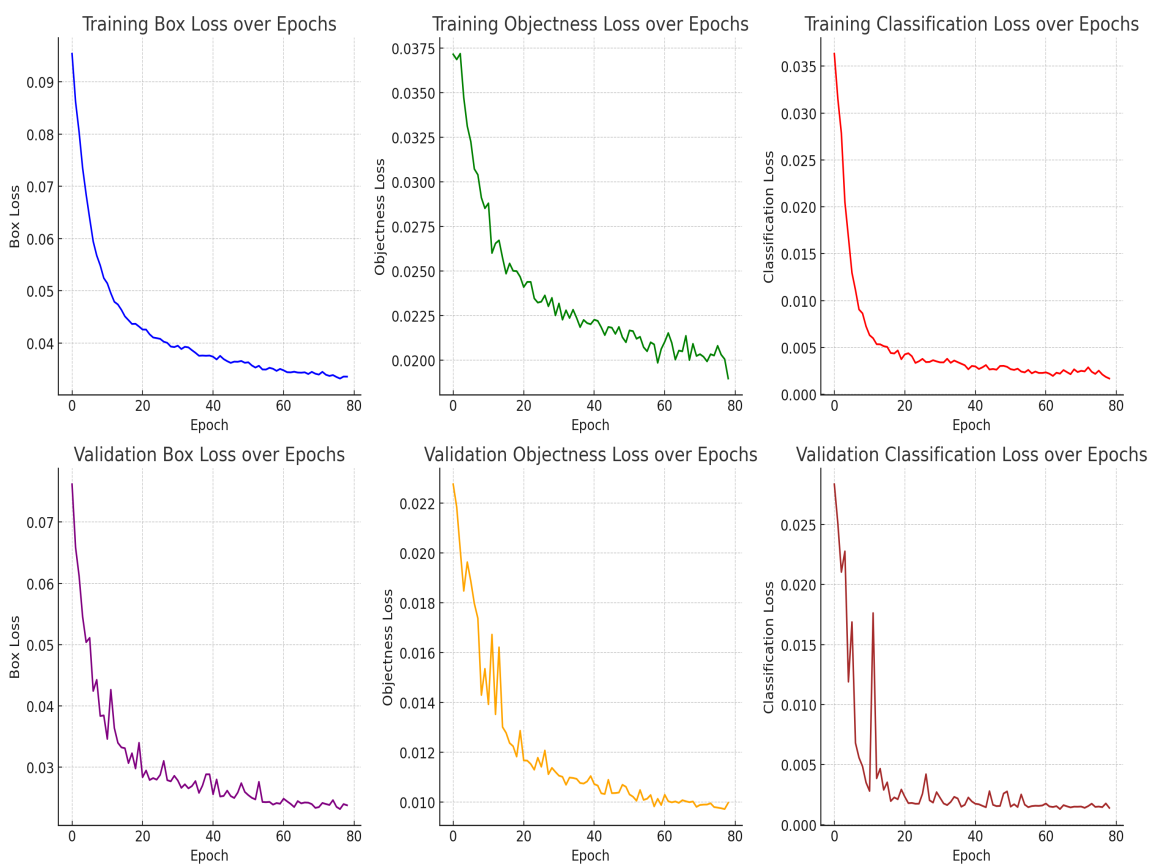


Figure 5.2: Trends In Losses Over Epochs

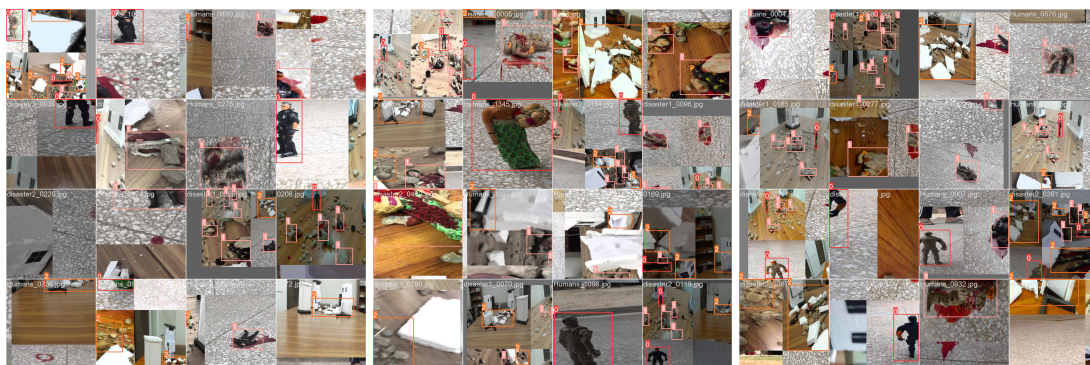


Figure 5.3: Object detection Results

5.3 Discussion

5.3.1 Effectiveness of Custom Dataset

The custom dataset, made with simulated disaster scenarios, was effective in training the YOLOv5 model. It included diverse images showing human miniatures in different positions and orientations, helping the model generalize well. The dataset had clear annotations for three classes: Alive, Injured, and Disaster, making it easier for the model to learn and identify these distinctions.

5.3.2 Model Performance

The model demonstrated a remarkable improvement in performance across the training epochs, as indicated by the recorded metrics: precision, recall, mAP 0.5, and mAP 0.5:0.95. By the conclusion of the training process, the model attained high precision and recall values, showcasing its capability to accurately detect and classify objects within the disaster environment.

5.3.3 Hardware-Software Integration

The project's success relied on the seamless integration of hardware components (Jetson Xavier, robotic arm, DC motors, lidar sensor, and camera) with the software stack (ROS Noetic, OpenCV, Python, and YOLOv5). Effective communication between these components enabled real-time object detection, obstacle avoidance, and precise movement control.

5.3.4 Real-Time Object Detection and Navigation

The YOLOv5 model's real-time detection allowed the robot to quickly respond to detected objects and navigate the environment efficiently. By combining object detection with SLAM (Simultaneous Localization and Mapping), the robot could avoid obstacles and concentrate on rescuing injured individuals.

5.3.5 Code For Project

The code for project is available at my GitHub repository. " <https://github.com/isupercoder400/Management-Robot> " use this link to get access to it.

5.4 Challenges and Limitations

5.4.1 Dataset Limitations

Although the custom dataset was effective, its limited size and the synthetic nature of the images might not fully represent real-world disaster scenarios. The diversity of real disasters can vary greatly, and the model may require additional training on more varied and realistic datasets to improve its robustness.

5.4.2 Hardware Constraints

While the computational power of the Jetson Xavier is sufficient for this project, it may fall short for more complex and larger-scale deployments. Handling higher-resolution images or more intricate scenes might necessitate more powerful hardware. Additionally, the robot's physical components, like the robotic arm and DC motors, have limitations in precision and strength, potentially impacting its ability to manage larger or more delicate objects

5.4.3 Real-Time Processing

Achieving real-time processing with high accuracy is challenging. It's crucial to balance processing speed and model accuracy, especially in dynamic and unpredictable environments. Latency in communication between components, particularly when managing large data streams from the camera and Lidar, can affect the robot's responsiveness.

5.4.4 Safety and Reliability

Ensuring the robot's safety and reliability in real disaster scenarios is critical. Any malfunction or incorrect detection could have severe consequences. Extensive testing and validation are required to ensure the robot's reliability before deployment in real-world situations.

5.5 Conclusion

In conclusion, the development of the disaster management robot showcased the practicality of using YOLOv5 and ROS Noetic for real-time object detection and navigation in disaster scenarios. The insights gained from this project lay a strong foundation for further research and development. Overcoming the identified challenges and limitations will be essential for advancing the technology and ensuring its effectiveness and reliability in real-world applications.

Chapter 6

Conclusion

6.1 Summary Of Findings

The development and deployment of the disaster management robot were valuable and gave a glimpse into the opportunity to combine better object detection algorithms with robots for disaster relief. The key findings from the project are summarized below:

6.1.1 Effective Object Detection

The custom-trained YOLOv5 model successfully detected objects within disaster scenarios, achieving high precision and recall. The use of a custom dataset with three classes (Alive, Injured, Disaster) enabled the model to effectively distinguish between different types of objects, crucial for targeted rescue operations.

6.1.2 Real-Time Performance

The robot performed impressively in real-time, quickly detecting objects and navigating around obstacles using SLAM and PID control algorithms. Thanks to YOLOv5's real-time object detection and effective control logic, the robot responded promptly and accurately to everything it detected.

6.1.3 Challenges and Limitations

The project faced several challenges, including dataset limitations, hardware constraints, real-time processing demands, and environmental factors. Addressing these challenges is essential for improving the robustness and reliability of the system in real-world applications.

6.2 Future Work

While doing the current project we have laid a strong foundation still several areas for future work and improvements have been identified.

6.2.1 Dataset Expansion And Hardware Enhancement

Expanding the dataset to include a wider variety of real-world disaster scenarios would improve the model's robustness and generalization. Collaborating with disaster response agencies to obtain real-world data and augmenting the dataset with synthetic data from various simulations could be beneficial. Upgrading the computational hardware to more powerful platforms, such as the latest NVIDIA GPUs, would allow for processing higher-resolution images and more complex scenes. Additionally, enhancing the robot's physical components, such as using more precise and stronger robotic arms, would improve its handling capabilities.

6.2.2 Advanced Algorithm And Autonomous Decision Making

Implementing more advanced algorithms for real-time processing, such as optimizing the YOLOv5 model for faster inference, would enhance the system's performance. Exploring other object detection models like SSD (Single Shot Detection) and integrating them with YOLOv5 could also provide complementary strengths. Advancing the autonomous decision-making capabilities of the robot by incorporating machine learning algorithms for predictive an-

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analytics and situational awareness would improve its effectiveness in disaster response. This could involve integrating reinforcement learning techniques to optimize the robot's actions based on real-time feedback.

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